Yuvashri Shanmugasundaram

M.Sc. Machine Design Student, KTH | Master's Thesis Candidate, Jan–Jun 2027 | Open to relocation
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PROFILE

I've always wanted to know why a design works, what makes it fail and how it can be made better, and that's what drew me to mechanical engineering. I'm now a Machine Design student at KTH, and my projects so far have taken me through mechanical design, CAD, CFD, vehicle dynamics, requirements engineering and root-cause analysis. Right now I'm helping build a test rig with Scania R&D, and a highlight of my bachelor's was leading the SAE BAJA team as its first female captain.

EDUCATION

M.Sc. Machine Design | Aug 2025 – Jun 2027
KTH Royal Institute of Technology, Stockholm, Sweden

- Courses: Component Design, Machine Dynamics, Testing Techniques in Solid Mechanics, Systems Engineering, Tribology

B.Eng. Automobile Engineering | Nov 2021 – Jul 2025
Madras Institute of Technology, Anna University, Chennai, India

- GPA 8.53/10, First Class with Distinction. Courses: Vehicle Control Systems, Dynamics of Ground Vehicles, Road Vehicle Aerodynamics, Machine Learning Using Python

PROJECTS

Modular Glare-Evaluation Test Rig for Truck Cabins – KTH and Scania R&D | Jan 2026 – Jan 2027

- Designing Solid Edge CAD and 3D-printed mounts in a 5-student team (6-DOF screen, 5-DOF mirror, lighting)
- Took the concept from functional decomposition to embodiment design
- Helping build the rig, with Raspberry Pi control and calibrated HDR glare measurement

Aerodynamic Optimisation of an EV Front End to Minimise Drag – B.Eng. Thesis | Aug 2024 – Apr 2025

- Modelled Hyundai Kona Electric front-end variants; the combined design cut drag coefficient by 17.83% in CFD
- Simulated in ANSYS; tested 3D-printed 1:20 models in a wind tunnel; manuscript under review for journal publication

Train-to-Wayside Communication (CBTC), Yellow Line – Systems Engineering, KTH | May 2026

- In a 6-person team, made safety-critical requirements verifiable in Polarion ALM (e.g. 100 ms latency, 90 s headway)

Eco-Design Analysis and Redesign of a Bicycle Dynamo – Eco Design, KTH | Feb 2026

- Led the 10 Golden Principles assessment; our redesign cut life-cycle CO2 emissions by over 64% in the team's LCA

EXPERIENCE

Student Intern | Jun 2024 – Jul 2024
Automotive Research Association of India (ARAI), Pune, India

- Case study on alternative fuels and dynamometer engine testing, instrumentation and emission analysers

Project Intern | Jan 2024 – Feb 2024
Combat Vehicles R&D Establishment (CVRDE), DRDO, Chennai, India

- Designed a photodiode-based Arduino circuit to detect oil dilution in a tank engine, verified in Tinkercad

Industrial Trainee | Jul 2023 – Aug 2023
Tractors and Farm Equipment Ltd (TAFE), India

- Traced bevel-gear tooth-root failures (Ishikawa); revised fillet radius and tooth thickness in CAD

LEADERSHIP & ACTIVITIES

Team Captain (2025) and Suspension Head (2023) – SAE BAJA India, Team MIT Racing | 2023 – 2025

- First female captain; led the team to the Cost Award (110 teams); department Best Captaincy Award
- As suspension head, ran roll and bump-steer studies in MSC ADAMS and Lotus SHARK; Design Event Top 10

Vehicle Dynamics Group – KTH Formula Student (DeV20) | Sep 2025 – Present

- Suspension modelling in MATLAB/Simulink and MSC ADAMS; attended FS Switzerland, FS Czech, Nordic Test Event

SAEINDIA Bicycle Design Challenge – 3rd place overall (finance lead; CAD and fabrication) | 2024

Converging Energy Technologies (CET) Conference, Oskarshamn – selected by KTH | 2026

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, CATIA V5, Solid Edge, ANSYS (CFD), MSC ADAMS, Lotus SHARK, MATLAB/Simulink
Software & Data: Python (incl. machine learning basics), Minitab, Power BI, SAP (basics), MS Office
Prototyping: 3D printing, Raspberry Pi, Arduino Uno, sensor selection, circuit simulation (Tinkercad)
Methods: Requirements management (Polarion ALM), concept development, morphological matrix, LCA, root-cause analysis
Languages: English (C1), Tamil (Mother tongue), Hindi (C1), Swedish (A1)
Interests: Running and workouts, singing, travelling and exploring new countries, reading